

Assignment 8: Time Series Analysis

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OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on generalized linear models.

Directions

1. Rename this file `<FirstLast>_A08_TimeSeries.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change “Student Name” on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Be sure to **answer the questions** in this assignment document.
5. When you have completed the assignment, **Knit** the text and code into a single PDF file.

Set up

1. Set up your session:
 - Check your working directory
 - Load the tidyverse, lubridate, zoo, and trend packages
 - Set your ggplot theme

```
getwd()
```

```
## [1] "/Users/mayike/Desktop/Duke/EDE_Fall2025"
```

```
library(tidyverse)
```

```
## Warning: package 'ggplot2' was built under R version 4.4.1
```

```
## Warning: package 'tidyr' was built under R version 4.4.1
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.5
## v forcats    1.0.0      v stringr    1.5.1
## v ggplot2    3.5.2      v tibble     3.2.1
## v lubridate  1.9.3      v tidyr      1.3.1
## v purrr      1.0.2
```

```
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag() masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(lubridate)
library(zoo)
```

```
## Warning: package 'zoo' was built under R version 4.4.1
```

```
##
## Attaching package: 'zoo'
##
## The following objects are masked from 'package:base':
##
## as.Date, as.Date.numeric
```

```
library(trend)
```

```
## Warning: package 'trend' was built under R version 4.4.1
```

2. Import the ten datasets from the Ozone_TimeSeries folder in the Raw data folder. These contain ozone concentrations at Garinger High School in North Carolina from 2010-2019 (the EPA air database only allows downloads for one year at a time). Import these either individually or in bulk and then combine them into a single dataframe named `GaringerOzone` of 3589 observation and 20 variables.

```
#1
EPA2010 <- read.csv("./Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2010_raw.csv", stringsAsFactors = T)
EPA2011 <- read.csv("./Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2011_raw.csv", stringsAsFactors = T)
EPA2012 <- read.csv("./Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2012_raw.csv", stringsAsFactors = T)
EPA2013 <- read.csv("./Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2013_raw.csv", stringsAsFactors = T)
EPA2014 <- read.csv("./Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2014_raw.csv", stringsAsFactors = T)
EPA2015 <- read.csv("./Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2015_raw.csv", stringsAsFactors = T)
EPA2016 <- read.csv("./Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2016_raw.csv", stringsAsFactors = T)
EPA2017 <- read.csv("./Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2017_raw.csv", stringsAsFactors = T)
EPA2018 <- read.csv("./Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2018_raw.csv", stringsAsFactors = T)
EPA2019 <- read.csv("./Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2019_raw.csv", stringsAsFactors = T)

#combine into one dataset
GaringerOzone <- rbind(EPA2010, EPA2011, EPA2012, EPA2013, EPA2014, EPA2015, EPA2016, EPA2017, EPA2018, EPA2019)
#write.csv(GaringerOzone, "./Data/Processed/EPAair_Garinger_combined.csv")
#GaringerOzone <- read.csv("./Data/Processed/EPAair_Garinger_combined.csv", stringsAsFactors = T)
#GaringerOzone <- GaringerOzone %>%
# select(-X)

#set theme
mytheme <- theme_classic(base_size = 14) +
  theme(axis.text = element_text(color = "black"),
        legend.position = "top")
theme_set(mytheme)
```

Wrangle

3. Set your date column as a date class.
4. Wrangle your dataset so that it only contains the columns Date, Daily.Max.8.hour.Ozone.Concentration, and DAILY_AQI_VALUE.
5. Notice there are a few days in each year that are missing ozone concentrations. We want to generate a daily dataset, so we will need to fill in any missing days with NA. Create a new data frame that contains a sequence of dates from 2010-01-01 to 2019-12-31 (hint: `as.data.frame(seq())`). Call this new data frame Days. Rename the column name in Days to "Date".
6. Use a `left_join` to combine the data frames. Specify the correct order of data frames within this function so that the final dimensions are 3652 rows and 3 columns. Call your combined data frame GaringerOzone.

```
# 3
GaringerOzone$Date <- mdy(GaringerOzone$Date)
# 4
GaringerOzone.daily <- GaringerOzone %>%
  select(Date, Daily.Max.8.hour.Ozone.Concentration, DAILY_AQI_VALUE)
# 5
Days <- as.data.frame(seq(as.Date("2010-01-01"),
                          as.Date("2019-12-31"),
                          1))
colnames(Days) <- c("Date")
# 6
GaringerOzone <- left_join(Days, GaringerOzone.daily)
```

```
## Joining with 'by = join_by(Date)'
```

```
summary(GaringerOzone$Daily.Max.8.hour.Ozone.Concentration) # 63 NAs in the dataframe
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.     NA's
## 0.00200 0.03200 0.04100 0.04163 0.05100 0.09300      63
```

Visualize

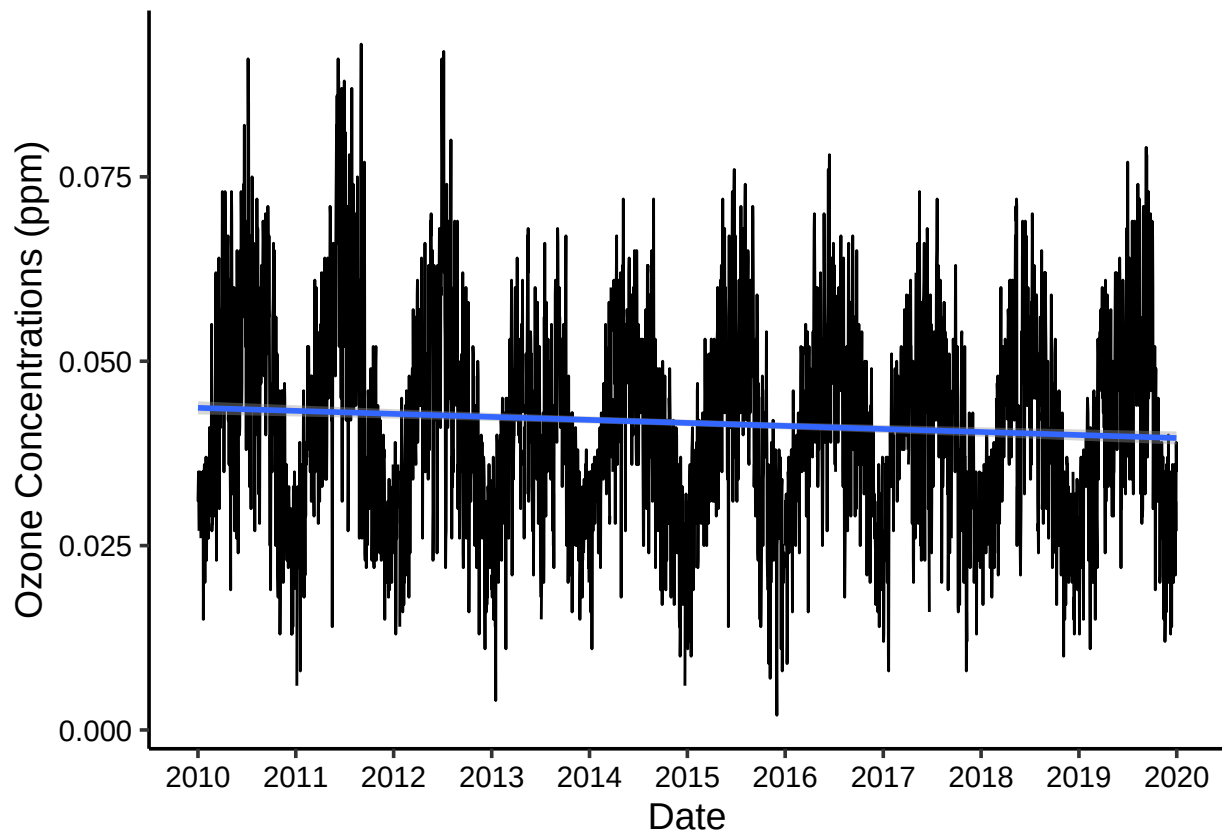
7. Create a line plot depicting ozone concentrations over time. In this case, we will plot actual concentrations in ppm, not AQI values. Format your axes accordingly. Add a smoothed line showing any linear trend of your data. Does your plot suggest a trend in ozone concentration over time?

```
#7
lineplot <-
  ggplot(GaringerOzone, aes(x = Date, y = Daily.Max.8.hour.Ozone.Concentration))+
  geom_line()+
  ylab("Ozone Concentrations (ppm)")+
  scale_x_date(date_breaks = "1 year", date_labels = "%Y")+
  geom_smooth(method = "lm")

lineplot
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```

```
## Warning: Removed 63 rows containing non-finite outside the scale range
## ('stat_smooth()').
```



Answer: The plot (linear model) suggests a seasonal change and a slight decrease in ozone concentrations over 10 years.

Time Series Analysis

Study question: Have ozone concentrations changed over the 2010s at this station?

8. Use a linear interpolation to fill in missing daily data for ozone concentration. Why didn't we use a piecewise constant or spline interpolation?

```
#8 No NAs in the new column Daily.Ozone
GaringerOzone <- GaringerOzone %>%
  mutate(Daily.Ozone = na.approx(Daily.Max.8.hour.Ozone.Concentration))
#No NAs in the new column Daily.Ozone
summary(GaringerOzone$Daily.Ozone) #0 NA
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00200 0.03200 0.04100 0.04151 0.05100 0.09300
```

Answer: Piecewise constant is unsuitable for data sets that vary continuously, and spline interpolation is used to create smooth curves, particularly when high accuracy is required. Here, linear

interpolation makes reasonable assumption when data changes overtime, and can better assist us to visualize the trend.

9. Create a new data frame called `GaringerOzone.monthly` that contains aggregated data: mean ozone concentrations for each month. In your pipe, you will need to first add columns for year and month to form the groupings. In a separate line of code, create a new Date column with each month-year combination being set as the first day of the month (this is for graphing purposes only)

```
#9
GaringerOzone.monthly <- GaringerOzone %>%
  mutate(year = year(Date), month = month(Date)) %>% #add new columns
  group_by(year, month) %>%
  summarise(mean = mean(Daily.Ozone)) %>%
  mutate(yearmonth = paste(as.factor(year), as.factor(month)))

## 'summarise()' has grouped output by 'year'. You can override using the
## '.groups' argument.
```

```
GaringerOzone.monthly$yearmonth <- ym(GaringerOzone.monthly$yearmonth)
```

10. Generate two time series objects. Name the first `GaringerOzone.daily.ts` and base it on the dataframe of daily observations. Name the second `GaringerOzone.monthly.ts` and base it on the monthly average ozone values. Be sure that each specifies the correct start and end dates and the frequency of the time series.

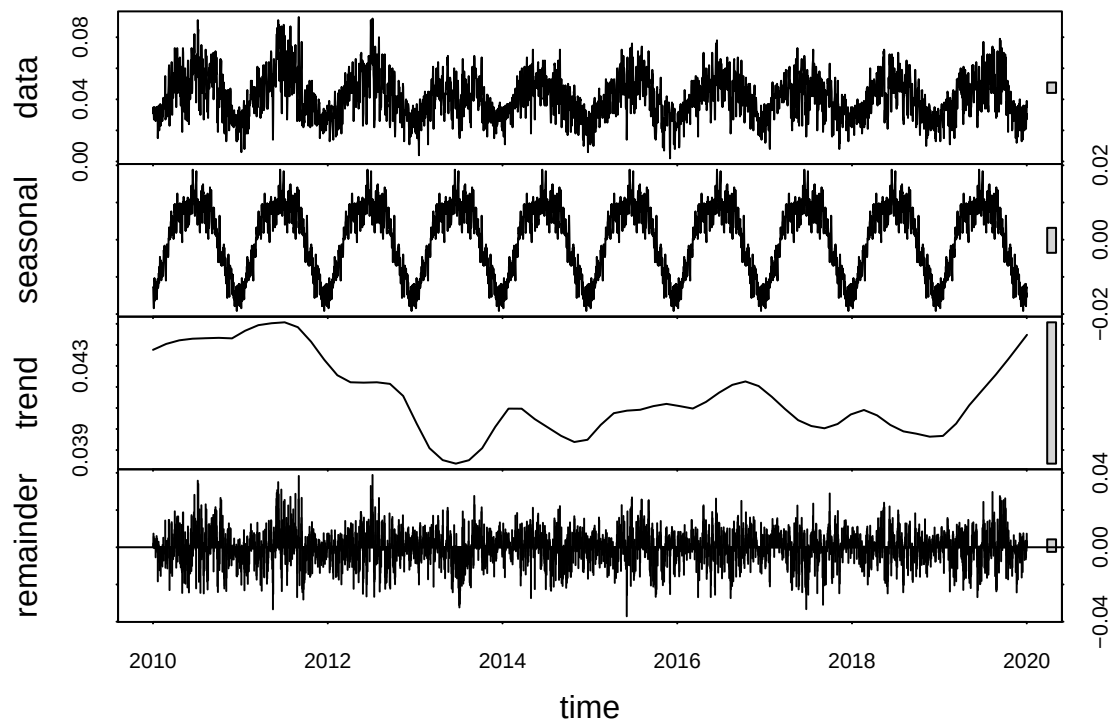
```
#10
fmonth <- month(first(GaringerOzone$Date))
fyear <- year(first(GaringerOzone$Date))

GaringerOzone.daily.ts <-
  ts(GaringerOzone$Daily.Ozone,
     start = c(fyear, fmonth),
     frequency = 365)

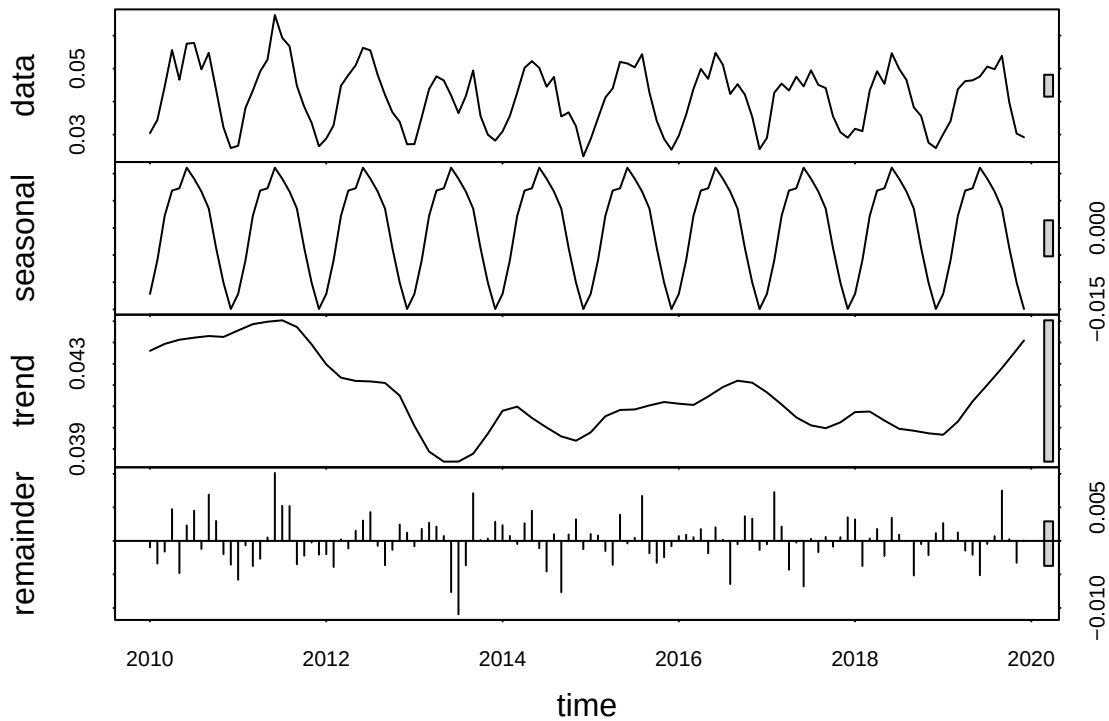
GaringerOzone.monthly.ts <-
  ts(GaringerOzone.monthly$mean,
     start = c(fyear, fmonth),
     frequency = 12)
```

11. Decompose the daily and the monthly time series objects and plot the components using the `plot()` function.

```
#11
GaringerOzone.daily.decomp <- stl(GaringerOzone.daily.ts, s.window = "periodic")
plot(GaringerOzone.daily.decomp)
```



```
GaringerOzone.month.decomp <- stl(GaringerOzone.monthly.ts, s.window = "periodic")
plot(GaringerOzone.month.decomp)
```



12. Run a monotonic trend analysis for the monthly Ozone series. In this case the seasonal Mann-Kendall is most appropriate; why is this?

```
#12
monthly.data.trend <- Kendall::SeasonalMannKendall(GaringerOzone.monthly.ts)
monthly.data.trend #p<0.5
```

```
## tau = -0.143, 2-sided pvalue =0.046724
```

```
summary(monthly.data.trend)
```

```
## Score = -77 , Var(Score) = 1499
## denominator = 539.4972
## tau = -0.143, 2-sided pvalue =0.046724
```

```
monthly.data.trend2 <- trend::smk.test(GaringerOzone.monthly.ts)
monthly.data.trend2
```

```
##
## Seasonal Mann-Kendall trend test (Hirsch-Slack test)
##
## data: GaringerOzone.monthly.ts
## z = -1.963, p-value = 0.04965
```

```
## alternative hypothesis: true S is not equal to 0
## sample estimates:
##      S varS
## -77 1499
```

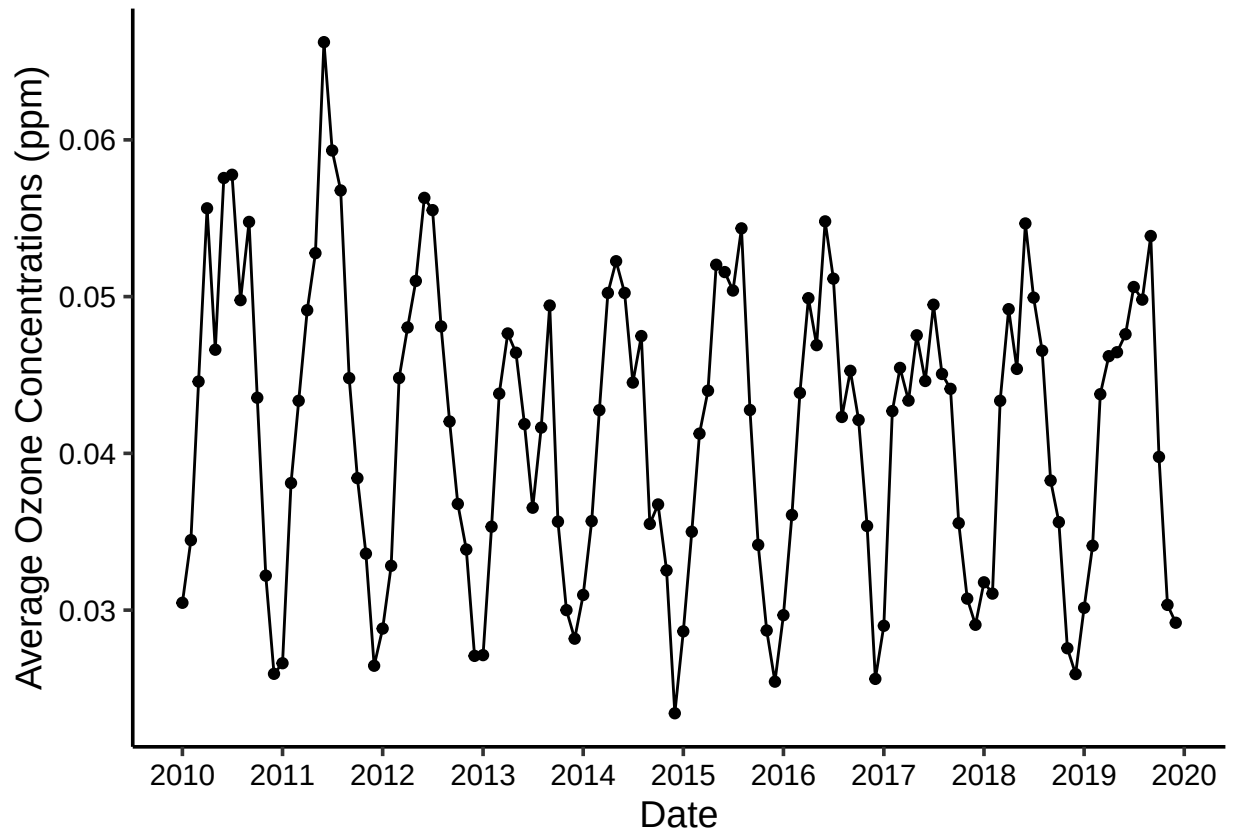
```
summary(monthly.data.trend2)
```

```
##
## Seasonal Mann-Kendall trend test (Hirsch-Slack test)
##
## data: GaringerOzone.monthly.ts
## alternative hypothesis: two.sided
##
## Statistics for individual seasons
##
## H0
##
##      S varS      tau      z Pr(>|z|)
## Season 1:  S = 0   15 125  0.333  1.252  0.21050
## Season 2:  S = 0   -1 125 -0.022  0.000  1.00000
## Season 3:  S = 0   -4 124 -0.090 -0.269  0.78762
## Season 4:  S = 0  -17 125 -0.378 -1.431  0.15241
## Season 5:  S = 0  -15 125 -0.333 -1.252  0.21050
## Season 6:  S = 0  -17 125 -0.378 -1.431  0.15241
## Season 7:  S = 0  -11 125 -0.244 -0.894  0.37109
## Season 8:  S = 0   -7 125 -0.156 -0.537  0.59151
## Season 9:  S = 0   -5 125 -0.111 -0.358  0.72051
## Season 10: S = 0  -13 125 -0.289 -1.073  0.28313
## Season 11: S = 0  -13 125 -0.289 -1.073  0.28313
## Season 12: S = 0   11 125  0.244  0.894  0.37109
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Answer: Because the data set exhibits seasonality, SMK can better analyze monotonic trend for seasonal data, while MannKendall is not suitable for seasonal data.

13. Create a plot depicting mean monthly ozone concentrations over time, with both a `geom_point` and a `geom_line` layer. Edit your axis labels accordingly.

```
# 13
ggplot(GaringerOzone.monthly, aes(x = yearmonth, y = mean))+
  geom_point()+
  geom_line()+
  xlab("Date")+
  scale_x_date(date_breaks = "1 year", date_labels = "%Y")+
  ylab("Average Ozone Concentrations (ppm)")
```

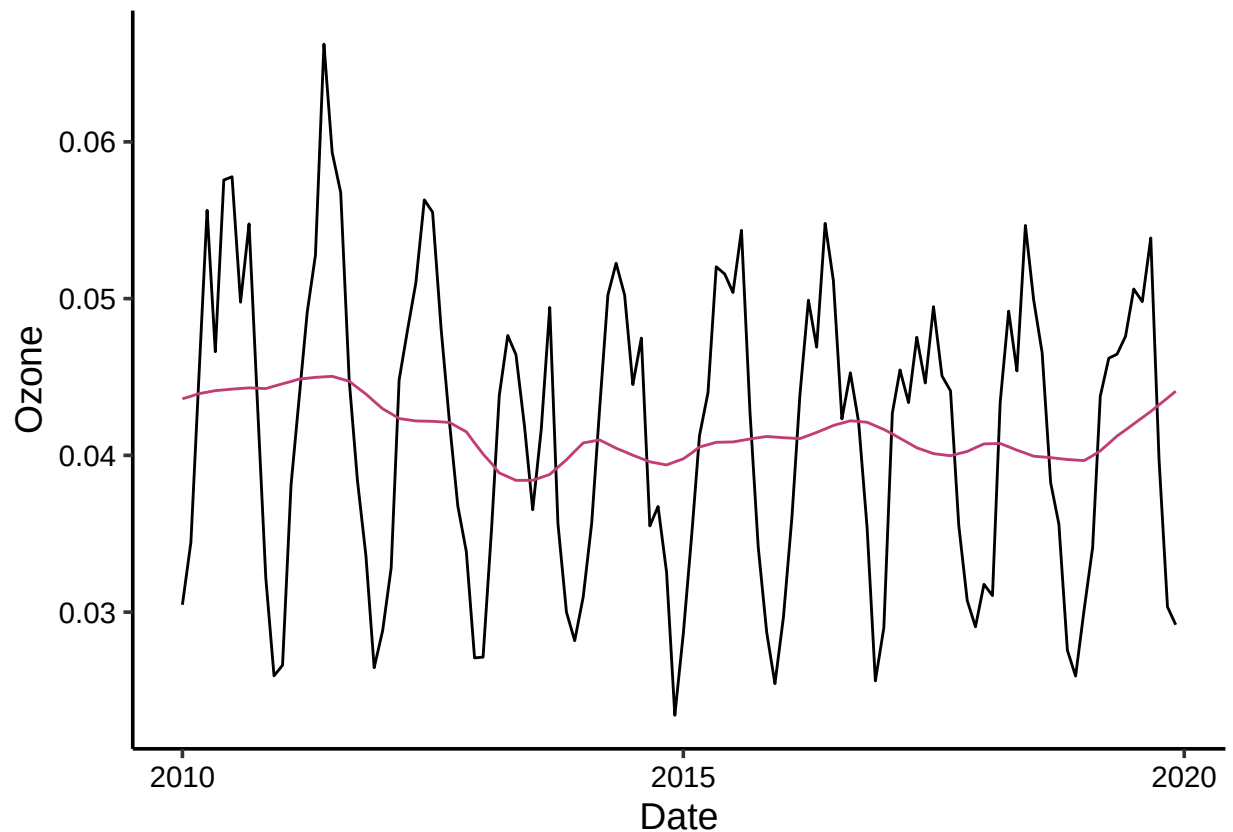
14. To accompany your graph, summarize your results in context of the research question. Include output from the statistical test in parentheses at the end of your sentence. Feel free to use multiple sentences in your interpretation.

Answer: Ozone concentrations are usually higher in summer with a maximum concentration of over 0.05 ppm each year, and decrease around winter. Ozone concentrations have an overall decreasing trend ($p=0.047 < 0.5$)

15. Subtract the seasonal component from the `GaringerOzone.monthly.ts`. Hint: Look at how we extracted the series components for the `EnoDischarge` on the lesson Rmd file.
16. Run the Mann Kendall test on the non-seasonal Ozone monthly series. Compare the results with the ones obtained with the Seasonal Mann Kendall on the complete series.

#15

```
monthly.component <- as.data.frame(GaringerOzone.month.decomp$time.series[,1:3])
# Visualize how the trend maps onto the data
monthly.component <-
  mutate(monthly.component,
    Ozone = GaringerOzone.monthly$mean,
    Date = GaringerOzone.monthly$yearmonth)
ggplot(monthly.component)+
  geom_line(aes(x=Date, y = Ozone))+
  geom_line(aes(x=Date, y = trend),color = "#c13d75ff")
```



#16

```
monthly.trend.mk <- Kendall::MannKendall(GaringerOzone.monthly.ts)
monthly.trend.mk #p = 0.337
```

```
## tau = -0.0594, 2-sided pvalue =0.33732
```

```
daily.trend.smk<- Kendall::SeasonalMannKendall(GaringerOzone.daily.ts)
daily.trend.smk #p = 0.0005
```

```
## tau = -0.0456, 2-sided pvalue =0.00051075
```

Answer: P value obtained from SMK on the complete series is 0.0005, indicating there is a trend even when seasonality is taken into consideration. While a much higher value ($p=0.337$) is obtained from Mann Kendall on monthly series, which is not as significant as treating it as seasonal data. (We have also obtained a 0.047 from SMK on monthly series in question 12.)